



DG Interns Hub

Project Title:

Final Cybersecurity Virtual Lab – Multi-Service
Deployment & Hardening

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Internship:

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Acknowledgement

I would like to express my sincere gratitude to **DG Interns Hub** for providing me the opportunity to work on this cybersecurity internship project.

This project has enhanced my practical understanding of cybersecurity concepts, server deployment, and system hardening. I am thankful to the entire training team for creating a structured and hands-on learning experience that helped me gain real-world exposure to network security and service configuration.

Introduction

Cybersecurity is one of the most crucial aspects of modern IT infrastructure, as organizations increasingly rely on digital services for communication, data storage, and business operations. To understand these concepts practically, this project focuses on building a secure virtual lab environment that includes the deployment of multiple network services and applying essential security hardening measures.

The main objective of this project is to create a controlled environment that simulates real-world enterprise systems such as Web Server, DNS Server, Mail Server, and FTP Server, and to secure them using industry-standard techniques like firewalls, access control, SSH hardening, Fail2Ban intrusion prevention, and secure configurations.

Through this project, I learned how servers interact in a network, how services are installed and configured, how to troubleshoot network issues, and how to strengthen systems against common cyber-attacks. This lab also helped me understand the role of network topology design, secure

communication protocols, and system monitoring in maintaining a reliable and secure IT infrastructure.

Purpose of the Project

The purpose of this project is to:

- Build a fully functional **multi-service network environment** using VirtualBox.
 - Deploy essential services such as the Web Server, DNS Server, Mail Server, and FTP Server.
 - Understand how these services work at the network and system level.
 - Apply **security best practices** including firewall rules, port management, SSH modification, user restrictions, and fail2ban.
 - Analyze server behavior using tools like **Nmap**, **nslookup**, **dig**, and system logs.
-

Services Deployed in This Project

The following services were installed, configured, tested, and secured:

1. **Web Server (Apache2 + HTTPS/SSL)**

Used for hosting web content and securing communication with SSL certificates.

2. **DNS Server (Bind9)**

Configured for domain name resolution and zone management.

3. **Mail Server (Postfix)**

Set up to send and receive local mails within the network.

4. **FTP Server (vsftpd)**

Deployed for secure file transfer with disabled anonymous login and user isolation.

Each service was assigned a specific IP address inside the virtual network:

- **Web Server:** 192.168.56.102
- **DNS Server:** 192.168.56.105
- **Mail Server:** 192.168.56.107
- **FTP Server:** 192.168.56.101

Importance of Cybersecurity & Hardening

Security hardening helps:

- Protect services from unauthorized access

- Prevent brute-force attacks
- Limit network exposure
- Strengthen server configurations
- Improve overall resilience against cyber-attacks

In this project, several key hardening techniques were implemented:

- **UFW firewall** configuration
 - **Fail2Ban** to prevent repeated login attacks
 - **SSH hardening** (disable root login, change port, allow specific users)
 - **FTP security** (disable anonymous login, enable user isolation)
 - **SSL encryption** for secure website communication
-

Lab Setup Overview

This cybersecurity virtual lab was created using Oracle VirtualBox to simulate a real-world network environment consisting of multiple servers and security components. Each virtual machine was configured with essential services, and all systems were interconnected using an internal network to provide isolated and controlled testing.

Virtualization Platform

- VirtualBox Version: Oracle VM VirtualBox 7.x
- Host Operating System: Windows 10 / 11 (64-bit)
- Network Mode Used: *Internal Network (intnet)* for communication between servers
- Optional NAT adapter added for updates

Guest Operating Systems (VMs Used)

VM Role	OS	IP Address	Purpose
Web Server	Kali Linux	192.168.56.102	Host Apache2 and HTTPS

VM Role	OS	IP Address	Purpose
DNS Server	Kali Linux	192.168.56.105	Host Bind9 and zone resolution
Mail Server	Kali Linux	192.168.56.107	Host Postfix for local mail transfer
FTP Server	Ubuntu	192.168.56.101	Host vsftpd for file transfer
Attacker / Test Machine	Kali Linux	DHCP / Manual	Used for Nmap & penetration testing

VM Hardware Configuration

Each Ubuntu server machine was provisioned with:

- **2 GB RAM**
- **2 CPU Cores**
- **20–25 GB Storage**
- **Internal Network Adapter**
- **Optional NAT Adapter for updates**

Tools Installed

Web Server:

- Apache2
- OpenSSL / Self-signed certificate or Let's Encrypt (optional)

DNS Server:

- Bind9
- DNS utilities (dig, nslookup, named-checkconf)

Mail Server:

- Postfix
- mailutils
- Log monitoring tools

FTP Server:

- vsftpd

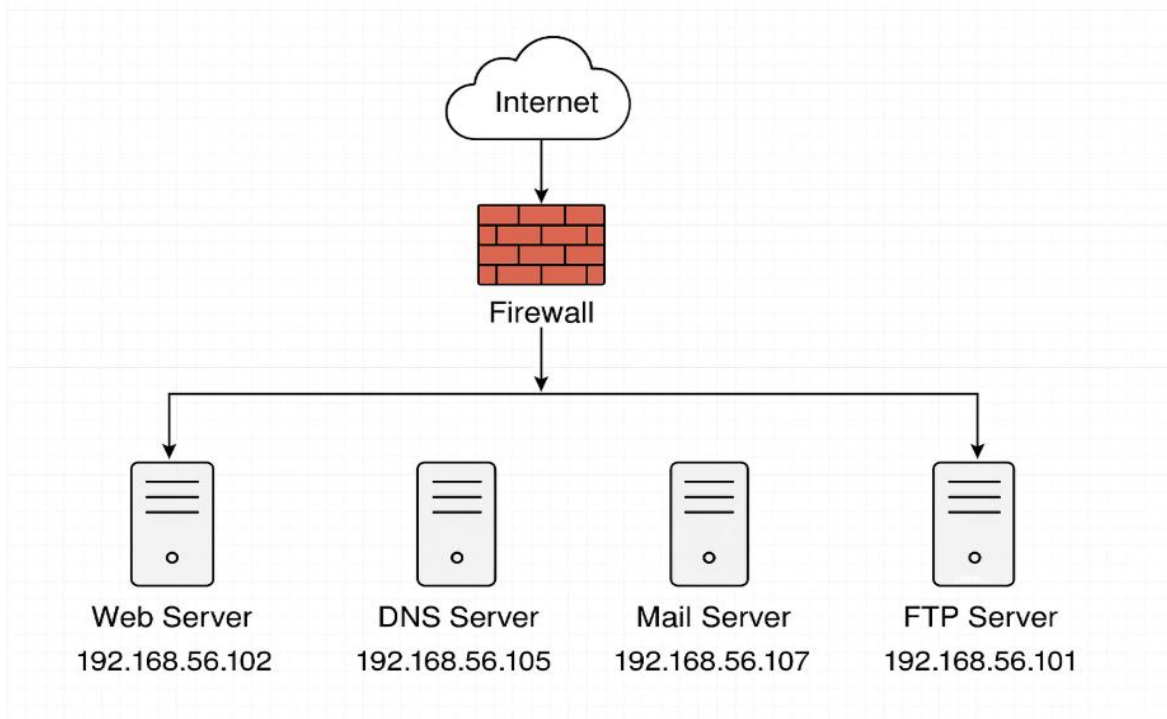
Security Tools:

- UFW Firewall
- Fail2Ban
- SSH hardening (OpenSSH)
- Nmap (for testing)

Network Topology Diagram

The virtual lab uses an internal network to interconnect all four servers.

Each service has a unique IP address assigned manually.



Description:

- All servers connect to a virtual switch (Internal Network).
- The firewall layer filters traffic.
- Each server runs one primary service.
- Kali Linux performs penetration testing.

Services Configured

1) Web Server (Apache2 + HTTPS)

1.1 Installation Steps

```
sudo apt update
```

```
sudo apt install apache2
```

1.2 Enable Firewall Rules

```
sudo ufw allow 80/tcp
```

```
sudo ufw allow 443/tcp
```

1.3 Create Self-Signed SSL Certificate

```
sudo openssl req -x509 -nodes -days 365 -newkey  
rsa:2048 \
```

```
-keyout /etc/ssl/private/apache.key \
```

```
-out /etc/ssl/certs/apache.crt
```

1.4 Configure Virtual Host

Edit the SSL config file:

```
sudo nano /etc/apache2/sites-available/default-ssl.conf
```

Set:

SSLEngine on

SSLCertificateFile /etc/ssl/certs/apache.crt

SSLCertificateKeyFile /etc/ssl/private/apache.key

Enable SSL Module:

```
sudo a2enmod ssl
```

```
sudo a2ensite default-ssl.conf
```

```
sudo systemctl restart apache2
```

1.5 HTTPS Testing

Using browser or:

```
curl -k https://192.168.56.102
```

Hardening Applied

- ✓ Disable directory listing
- ✓ Enable firewall rules
- ✓ Disable default Apache pages
- ✓ Restrict access logs

Nmap Scan

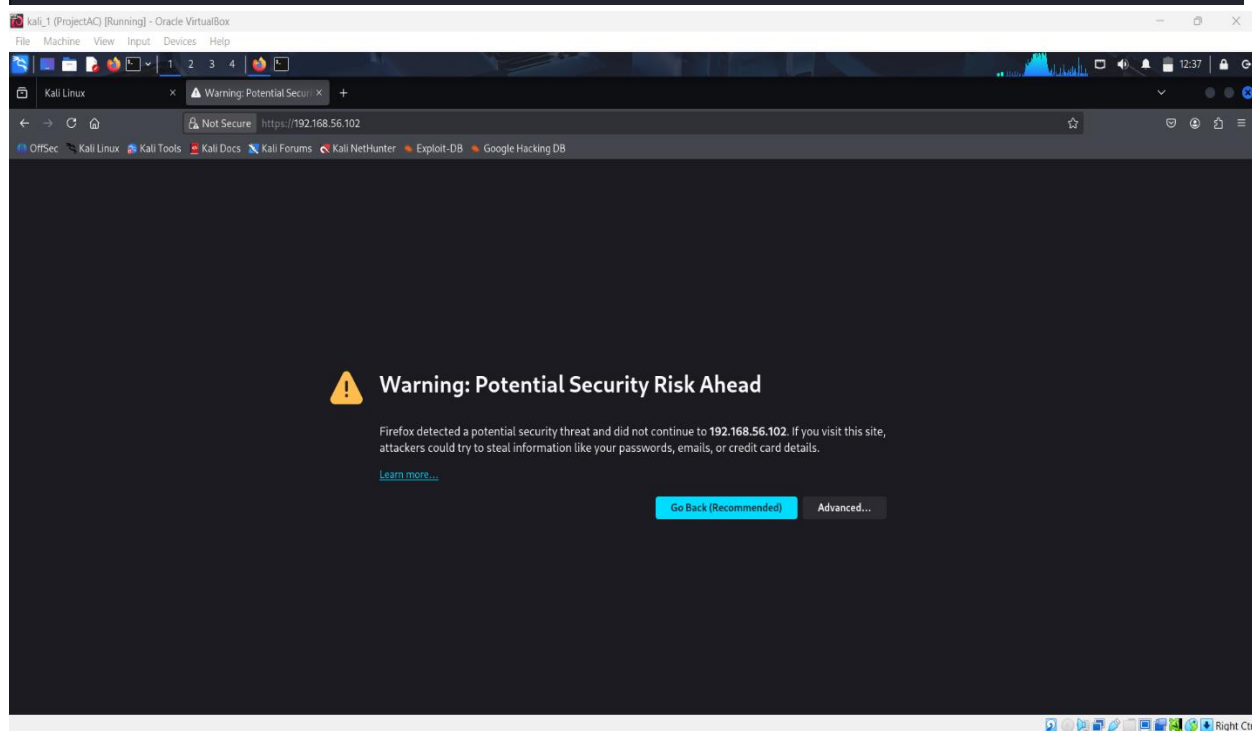
nmap -sV 192.168.56.102

Attach Screenshots:

```
kali (main) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali:~$ sudo apt install apache2 -y
apache2 is already the newest version (2.4.65-3+b1).
apache2 set to manually installed.
The following packages were automatically installed and are no longer required:
amass-common          libglvnd-dev          libpython3.12t64      python3-gpg
crackmapexec          libgtksourceview-3.0-1 libqt5ct-common1.8    python3-kismetcapturebtgeiger
firebird3.0-common    libgtksourceview-3.0-common libqt5sensors5        python3-kismetcapturefreaklabszigbee
firebird3.0-common-doc libgtksourceviewmm-3.0-0v5 libqt5webkit5         python3-kismetcapturertl433
firmware-ti-connectivity libgumbo2             libqt5x11extras5     python3-kismetcapturertladsb
icu-devtools          libhdf4-0-alt         librav1e0.7          python3-kismetcapturertlamr
imagemagick-6.q16     libhdf5-103-1t64     libsframe1           python3-nfsclient
libabsl20230802       libhdf5-hl-100t64    libsigsegv2          python3-ntlm-auth
libbfi01              libicu-dev           libsoup-2.4-1        python3-packaging-whl
libbluray2            libjxl0.9            libsoup2.4-common    python3-poetry-dynamic-versioning
libbson-1.0-0t64      liblbfgsb0           libsuperlu6          python3-pywebview
libc++1-19            libldap-2.5-0        libtag1v5            python3-requests-ntlm
libc++abi1-19         libmagickcore-6.q16-7-extra libtag1v5-vanilla    python3-tomlkit
libcapstone4          libmagickcore-6.q16-7t64 libtagc0             python3-wheel-whl
libconfig++9v5        libmagickwand-6.q16-7t64 libtheora0           python3-zombie-imp
libconfign9           libmbedcrypto7t64    libudfread0         python3.12
libdirectfb-1.7-7t64  libmongoc-1.0-0t64   libunwind-19        python3.12-dev
libegl-dev            libmongocrypt0       libutempter0        python3.12-minimal
libflac12t64         libmsgpack-0-1       libvpx9             python3.12-tk
libfmt9              libnet1              libwebrtc-audio-processing1 python3.12-venv
libfuse3-3           libnetcdf19t64       libx264-164         ruby-zeitwerk
```

```
(kali㉿kali)-[~]  
$ sudo a2enmod ssl  
sudo a2ensite default-ssl.conf  
sudo systemctl restart apache2
```

```
Considering dependency mime for ssl:  
Module mime already enabled  
Considering dependency socache_shmcb for ssl:  
Module socache_shmcb already enabled  
Module ssl already enabled  
Site default-ssl already enabled
```



```
(kali㉿kali)-[~]  
$ nmap -p 443 192.168.56.102  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-06 12:39 EST  
Nmap scan report for 192.168.56.102  
Host is up (0.000078s latency).  
  
PORT      STATE SERVICE  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

2) DNS Server (Bind9)

2.1 Install Bind9

```
sudo apt install bind9 bind9utils bind9-doc
```

2.2 Configure named.conf.local

Add zone:

```
zone "mydomain.local" {  
    type master;  
    file "/etc/bind/db.mydomain.local";  
};
```

2.3 Create Zone File

Path:

```
/etc/bind/db.mydomain.local
```

Example zone:

```
@ IN SOA ns1.mydomain.local. admin.mydomain.local. (  
    2 ; Serial  
    604800 ; Refresh
```

86400 ; Retry

2419200 ; Expire

604800) ; Negative Cache TTL

;

@ IN NS ns1

ns1 IN A 192.168.56.105

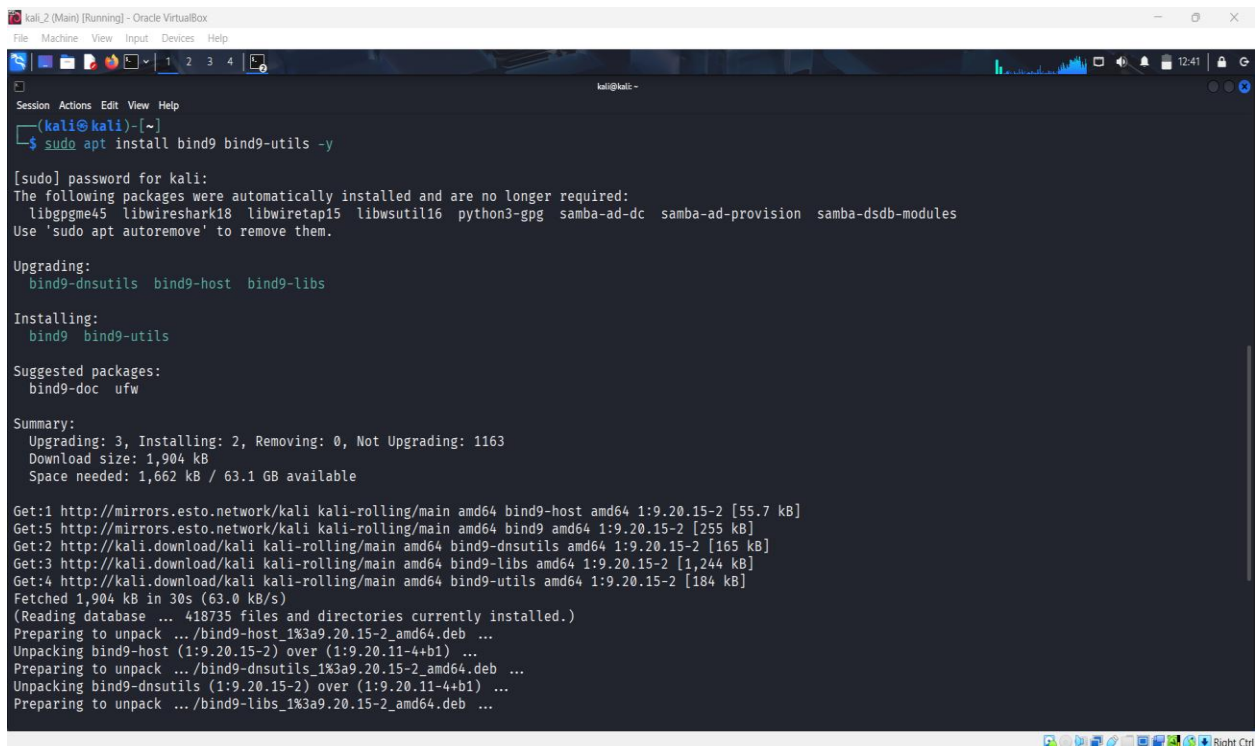
www IN A 192.168.56.102

2.4 DNS Resolution Testing

nslookup www.mydomain.local

dig ftp.mydomain.local

Screenshots:



```
kali_2 (Main) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
$ sudo apt install bind9 bind9-utils -y

[sudo] password for kali:
The following packages were automatically installed and are no longer required:
  libpgpme45 libwireshark18 libwiretap15 libwsutil16 python3-gpg samba-ad-dc samba-ad-provision samba-dsdb-modules
Use 'sudo apt autoremove' to remove them.

Upgrading:
  bind9-dnsutils bind9-host bind9-libs

Installing:
  bind9 bind9-utils

Suggested packages:
  bind9-doc ufw

Summary:
  Upgrading: 3, Installing: 2, Removing: 0, Not Upgrading: 1163
  Download size: 1,904 kB
  Space needed: 1,662 kB / 63.1 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 bind9-host amd64 1:9.20.15-2 [55.7 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/main amd64 bind9 amd64 1:9.20.15-2 [255 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 bind9-dnsutils amd64 1:9.20.15-2 [165 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 bind9-libs amd64 1:9.20.15-2 [1,244 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 bind9-utils amd64 1:9.20.15-2 [184 kB]
Fetched 1,904 kB in 30s (63.0 kB/s)
(Reading database ... 418735 files and directories currently installed.)
Preparing to unpack .../bind9-host_1%3a9.20.15-2_amd64.deb ...
Unpacking bind9-host (1:9.20.15-2) over (1:9.20.11-4+b1) ...
Preparing to unpack .../bind9-dnsutils_1%3a9.20.15-2_amd64.deb ...
Unpacking bind9-dnsutils (1:9.20.15-2) over (1:9.20.11-4+b1) ...
Preparing to unpack .../bind9-libs_1%3a9.20.15-2_amd64.deb ...
```


kali_2 (Main) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Session Actions Edit View Help

GNU nano 8.6 /etc/bind/named.conf.local

```
//  
// Do any local configuration here  
//  
zone "mydomain.local" {  
    type master;  
    file "/etc/bind/db.mydomain.local";  
};
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

kali_2 (Main) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Session Actions Edit View Help

GNU nano 8.6 /etc/bind/db.mydomain.local

```
$TTL 86400  
@ IN SOA ns1.mydomain.local. admin.mydomain.local. (  
    2025120601 ; Serial  
    3600 ; Refresh  
    1800 ; Retry  
    604800 ; Expire  
    86400 ) ; Negative Cache TTL  
  
; Name servers  
@ IN NS ns1.mydomain.local.  
  
; A records  
ns1 IN A 192.168.56.105  
www IN A 192.168.56.105  
@ IN A 192.168.56.105
```

[Wrote 16 lines]

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

```
(kali㉿kali)-[~]
```

```
$ nslookup www.mydomain.local 192.168.56.105
```

```
Server:                192.168.56.105
Address:                192.168.56.105#53
```

```
Name:   www.mydomain.local
Address: 192.168.56.105
```

```
kali_1 (ProjectAC) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali -
Session Actions Edit View Help

(kali㉿kali)-[~]
$ dig www.mydomain.local

; <<>> DiG 9.20.15-2-Debian <<>> www.mydomain.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 65471
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 8be8bd55815d1a310100000069346f0bea787efeff88199c (good)
;; QUESTION SECTION:
;www.mydomain.local.      IN      A

;; ANSWER SECTION:
www.mydomain.local.      604800 IN      A      192.168.56.102

;; Query time: 0 msec
;; SERVER: 192.168.56.102#53(192.168.56.102) (UDP)
;; WHEN: Sat Dec 06 12:59:39 EST 2025
;; MSG SIZE rcvd: 91

(kali㉿kali)-[~]
```

3) Mail Server (Postfix)

3.1 Installation

```
sudo apt install postfix mailutils
```

Select: **Local Only**

3.2 Configuration File

```
/etc/postfix/main.cf
```

Set:

```
myhostname = mail.mydomain.local
```

```
mydestination = $myhostname, localhost.localdomain,  
localhost
```

3.3 Send Test Mail

```
echo "Hello Maildir" | mail -s "Test Mail" user1@mydomain.local
```

3.4 Logs

```
sudo tail -f /var/log/mail.log
```

Screenshots

```
kali_3 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo apt install postfix mailutils -y

Installing:
  mailutils postfix

Installing dependencies:
  gsasl-common libgsasl18 libkyotocabinet16v5 libmu-dbm9t64 libtlsrpt0
  guile-3.0-libs libgssglue1 libmailutils9t64 libntlm0 mailutils-common

Suggested packages:
  mailutils-mh postfix-doc postfix-mta-sts-resolver postfix-pcre procmail | dovecot-common
  mailutils-doc postfix-ldap postfix-mongodb postfix-pgsql resolvconf ufw
  postfix-cdb postfix-lmdb postfix-mysql postfix-sqlite sasl2-bin

Summary:
  Upgrading: 0, Installing: 12, Removing: 0, Not Upgrading: 1311
  Download size: 8,172 kB / 11.8 MB
  Space needed: 69.4 MB / 63.5 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 guile-3.0-libs amd64 3.0.10+really3.0.10-6 [6,884 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libgssglue1 amd64 0.9-2 [20.6 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libkyotocabinet16v5 amd64 1.2.80-2+b1 [316 kB]
Get:4 http://mirrors.esto.network/kali kali-rolling/main amd64 libmailutils9t64 amd64 1:3.20-3 [951 kB]
Fetched 2,853 kB in 8s (341 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libtlsrpt0:amd64.
```

```
(kali@kali)-[~]
$ sudo systemctl restart postfix
$ sudo systemctl status postfix

● postfix.service - Postfix Mail Transport Agent (main/default instance)
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; preset: disabled)
   Active: active (running) since Sat 2025-12-06 13:14:25 EST; 32ms ago
 Invocation: 1ebb266b606a43419f125cddb924b01
    Docs: man:postfix(1)
   Process: 7571 ExecStartPre=postfix check (code=exited, status=0/SUCCESS)
   Process: 7676 ExecStart=postfix debian-systemd-start (code=exited, status=0/SUCCESS)
  Main PID: 7684 (master)
    Tasks: 3 (limit: 2208)
   Memory: 3M (peak: 3.3M)
      CPU: 398ms
   CGroup: /system.slice/postfix.service
           └─7684 /usr/lib/postfix/sbin/master -w
             └─7685 pickup -l -t unix -u -c
               └─7686 qmgr -l -t unix -u

Dec 06 13:14:24 kali systemd[1]: Starting postfix.service - Postfix Mail Transport Agent (main/default instance)...
Dec 06 13:14:25 kali postfix/master[7684]: daemon started -- version 3.10.6, configuration /etc/postfix
Dec 06 13:14:25 kali systemd[1]: Started postfix.service - Postfix Mail Transport Agent (main/default instance).
```

```
(kali㉿kali)-[~]  
$ echo "Hello Maildir" | mail -s "Test Mail" user1@mydomain.local
```

```
(kali㉿kali)-[~]  
$ su - user1  
Password:  
(user1㉿kali)-[~]  
$ cat ~/Maildir/new/*  
Return-Path: <kali@kali>  
X-Original-To: user1@mydomain.local  
Delivered-To: user1@mydomain.local  
Received: by kali.mydomain.local (Postfix, from userid 1000)  
        id 2B5192C36EC; Sat, 06 Dec 2025 13:14:34 -0500 (EST)  
Subject: Test Mail  
To: <user1@mydomain.local>  
User-Agent: mail (GNU Mailutils 3.20)  
Date: Sat, 6 Dec 2025 13:14:34 -0500  
Message-Id: <20251206181434.2B5192C36EC@kali.mydomain.local>
```

4) FTP Server (vsftpd)

4.1 Installation

```
sudo apt install vsftpd
```

4.2 Important Configurations

Edit:

```
/etc/vsftpd.conf
```

Set:

```
anonymous_enable=NO
```

```
local_enable=YES
```

```
write_enable=YES
```

```
chroot_local_user=YES
```

Restart:

```
sudo systemctl restart vsftpd
```

4.3 FTP Login Test

[ftp 192.168.56.101](ftp://192.168.56.101)

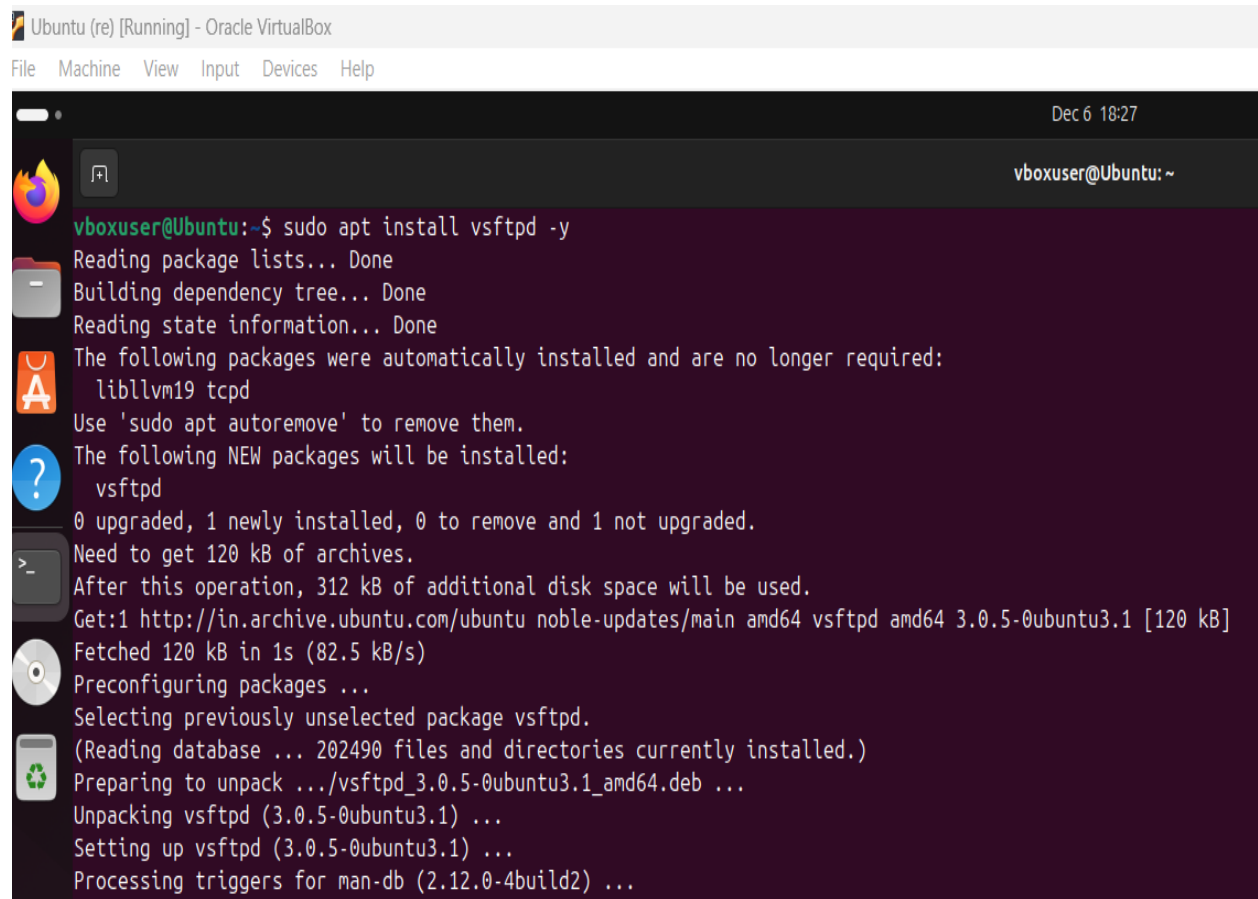
4.4 Upload / Download Test

Commands:

put file.txt

get file.txt

Screenshots



The screenshot shows a terminal window titled "Ubuntu (re) [Running] - Oracle VirtualBox". The terminal output shows the command `sudo apt install vsftpd -y` being executed. The output indicates that the package lists, dependency tree, and state information were read successfully. It then lists the packages that were automatically installed and are no longer required: `libllvm19` and `tcpd`. It also shows the new packages to be installed: `vsftpd`. The terminal output continues with the disk space requirements, the source of the package, and the installation progress.

```
Ubuntu (re) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@Ubuntu: ~
vboxuser@Ubuntu:~$ sudo apt install vsftpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libllvm19 tcpd
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  vsftpd
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 120 kB of archives.
After this operation, 312 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubuntu3.1 [120 kB]
Fetched 120 kB in 1s (82.5 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database ... 202490 files and directories currently installed.)
Preparing to unpack .../vsftpd_3.0.5-0ubuntu3.1_amd64.deb ...
Unpacking vsftpd (3.0.5-0ubuntu3.1) ...
Setting up vsftpd (3.0.5-0ubuntu3.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```
vboxuser@Ubuntu:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: enabled)
   Active: active (running) since Sat 2025-12-06 18:29:57 UTC; 9s ago
     Process: 4293 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
    Main PID: 4295 (vsftpd)
      Tasks: 1 (limit: 3468)
     Memory: 700.0K (peak: 1.5M)
        CPU: 15ms
      CGroup: /system.slice/vsftpd.service
              └─4295 /usr/sbin/vsftpd /etc/vsftpd.conf

Dec 06 18:29:57 Ubuntu systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Dec 06 18:29:57 Ubuntu systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

```
GNU nano 7.2 /etc/vsftpd.conf
# capabilities.
#
# Run standalone? vsftpd can run either from an inetd or as a standalone
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (:::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
# if your users expect that (022 is used by most other ftpd's)
local_umask=022
#
# Uncomment this to allow the anonymous FTP user to upload files. This only
# has an effect if the above global write enable is activated. Also, you will
# obviously need to create a directory writable by the FTP user.
#anon_upload_enable=YES
#
# Uncomment this if you want the anonymous FTP user to be able to create
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward


```
(kali㉿kali)-[~]  
$ ftp 192.168.56.101  
Connected to 192.168.56.101.  
220 (vsFTPd 3.0.5)  
Name (192.168.56.101:kali): vboxuser  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> put test.txt  
local: test.txt remote: test.txt  
229 Entering Extended Passive Mode (|||60784|)  
150 Ok to send data.  
0 0.00 KiB/s  
226 Transfer complete.  
ftp> ls  
229 Entering Extended Passive Mode (|||36731|)  
150 Here comes the directory listing.  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Desktop  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Documents  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Downloads  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Music  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Pictures  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Public  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Templates  
drwxr-xr-x 2 1000 1000 4096 Nov 08 06:11 Videos  
drwx----- 5 1000 1000 4096 Nov 19 10:24 snap  
-rw-rw-r-- 1 1000 1000 0 Dec 06 18:33 test.txt  
226 Directory send OK.  
ftp> get test.txt  
local: test.txt remote: test.txt  
229 Entering Extended Passive Mode (|||9098|)  
150 Opening BINARY mode data connection for test.txt (0 bytes).  
0 0.00 KiB/s  
226 Transfer complete.  
ftp> █
```

Security Hardening Applied

✓ UFW Firewall

- Allowed required ports: **22, 80, 443, 53, 25, 21**
- Denied everything else:

sudo ufw default deny incoming

sudo ufw default allow outgoing

```
(kali㉿kali)-[~]
└─$ sudo apt install ufw -y
sudo ufw allow 22
sudo ufw allow 80
sudo ufw allow 443
sudo ufw allow 53
sudo ufw allow 21
sudo ufw enable
sudo ufw status

ufw is already the newest version (0.36.2-9).
The following packages were automatically installed and are no longer required:
  libgpgme45 python3-gpg samba-ad-dc samba-ad-provision samba-dsdb-modules
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1156
Rule added
Rule added (v6)
Rule updated
Rule updated (v6)
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Firewall is active and enabled on system startup
Status: active
```

To	Action	From
--		
80	ALLOW	Anywhere
22	ALLOW	Anywhere
443	ALLOW	Anywhere
53	ALLOW	Anywhere
21	ALLOW	Anywhere
80 (v6)	ALLOW	Anywhere (v6)
22 (v6)	ALLOW	Anywhere (v6)
443 (v6)	ALLOW	Anywhere (v6)
53 (v6)	ALLOW	Anywhere (v6)

Fail2ban

Installation:

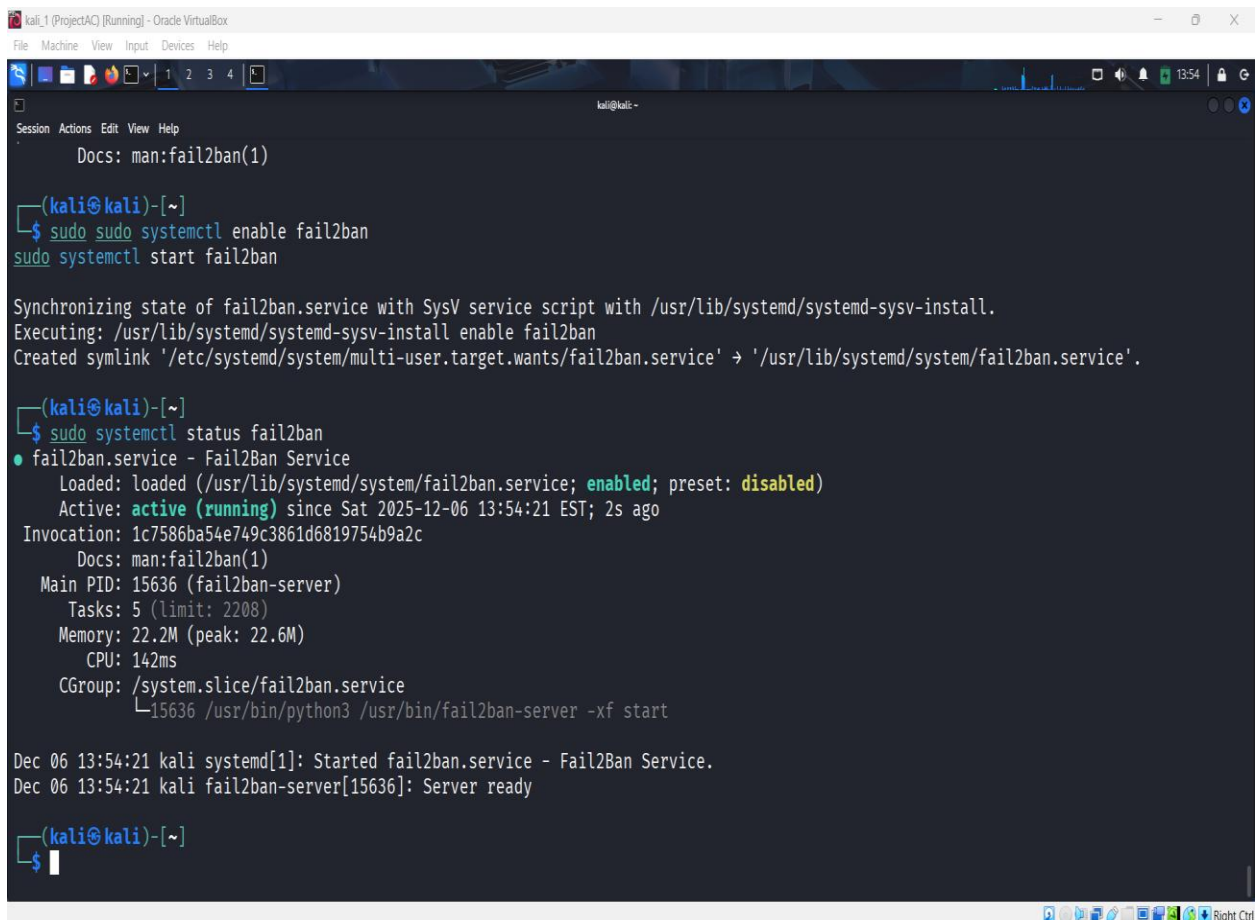
`sudo apt install fail2ban`

Configured jail:

`/etc/fail2ban/jail.local`

Enabled:

`sudo systemctl enable --now fail2ban`



```
kali_1 (ProjectAC) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Docs: man:fail2ban(1)

(kali@kali)-[~]
$ sudo systemctl enable fail2ban
sudo systemctl start fail2ban

Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' -> '/usr/lib/systemd/system/fail2ban.service'.

(kali@kali)-[~]
$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-12-06 13:54:21 EST; 2s ago
 Invocation: 1c7586ba54e749c3861d6819754b9a2c
    Docs: man:fail2ban(1)
  Main PID: 15636 (fail2ban-server)
    Tasks: 5 (limit: 2208)
  Memory: 22.2M (peak: 22.6M)
    CPU: 142ms
   CGroup: /system.slice/fail2ban.service
           └─15636 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Dec 06 13:54:21 kali systemd[1]: Started fail2ban.service - Fail2Ban Service.
Dec 06 13:54:21 kali fail2ban-server[15636]: Server ready

(kali@kali)-[~]
$
```

SSH Hardening

File:

/etc/ssh/sshd_config

Set:

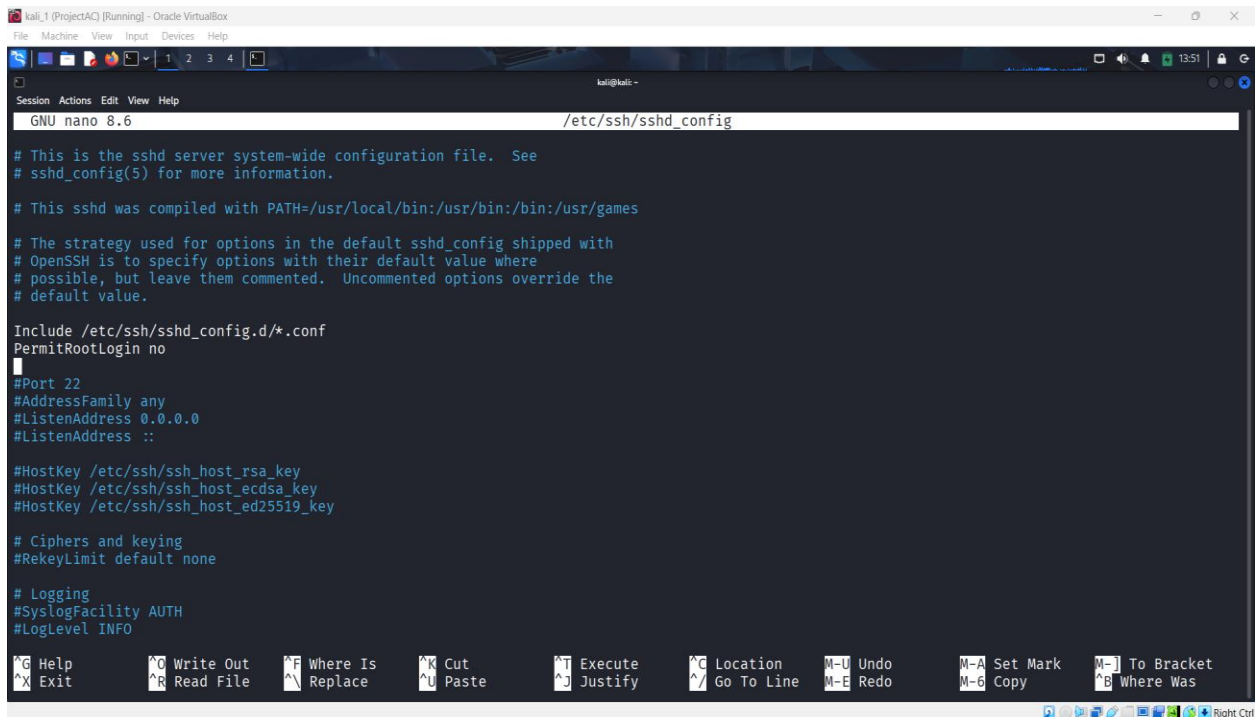
PermitRootLogin no

PasswordAuthentication yes/no

AllowUsers youruser

Restart SSH:

sudo systemctl restart ssh



The screenshot shows a terminal window titled 'kali_1 (ProjectAC) [Running] - Oracle VirtualBox'. The terminal is running the GNU nano 8.6 editor, editing the file /etc/ssh/sshd_config. The file content is as follows:

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf
PermitRootLogin no

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

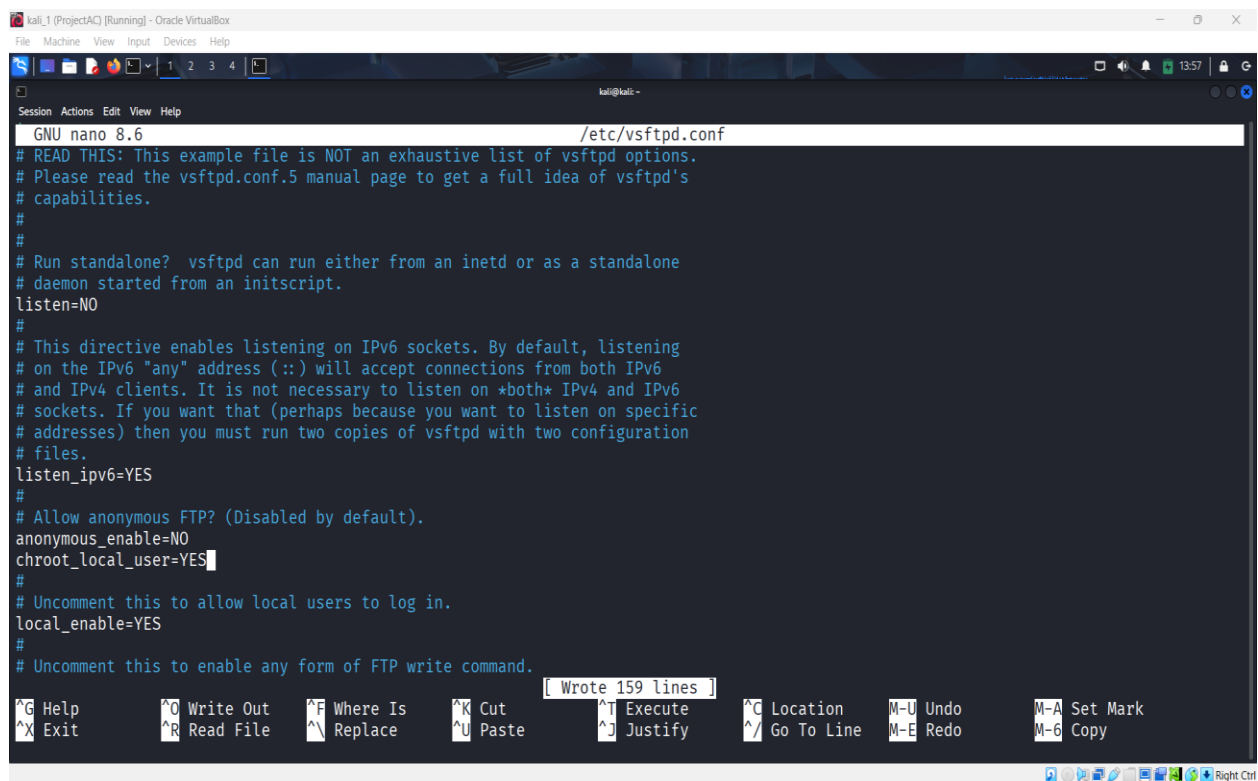
# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO
```

The terminal window also displays a help menu at the bottom with various keyboard shortcuts for nano editor operations.

FTP Hardening

- Disabled Anonymous Login
- chroot jail enabled
- Strong password rules
- UFW firewall restricted



The screenshot shows a Kali Linux virtual machine window titled "kali_1 (ProjectAC) [Running] - Oracle VirtualBox". Inside the VM, a terminal window is open, displaying the contents of the `/etc/vsftpd.conf` file using the `nano` text editor. The file contains various configuration options for vsftpd, including `listen=NO`, `listen_ipv6=YES`, `anonymous_enable=NO`, `chroot_local_user=YES`, and `local_enable=YES`. The terminal window has a status bar at the bottom showing "Wrote 159 lines" and a menu bar with various keyboard shortcuts like `^G Help`, `^O Write Out`, `^F Where Is`, `^K Cut`, `^T Execute`, `^C Location`, `M-U Undo`, `M-A Set Mark`, `^X Exit`, `^R Read File`, `^N Replace`, `^U Paste`, `^J Justify`, `^_ Go To Line`, `M-E Redo`, and `M-B Copy`.

```
GNU nano 8.6 /etc/vsftpd.conf
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
# Run standalone? vsftpd can run either from an inetd or as a standalone
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (:::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
chroot_local_user=YES
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
```

Challenges Faced & Solutions

1. DNS resolution not working

Reason: Misconfigured zone file

Fix: Corrected A records and SOA entry

2. UFW blocked SSH accidentally

Fix: Logged in through VirtualBox console and allowed SSH manually

3. FTP permission issues

Fix: Set correct ownership and enabled write permissions

4. Postfix mail stuck in queue

Fix: Started postfix service and fixed hostname configuration

Conclusion

This project provided hands-on experience in deploying and securing multiple network services in a virtualized environment. I learned how servers communicate, how DNS and mail routing work, how to secure communication with SSL, and how to protect services using firewall and intrusion prevention tools. The lab reflects real-world system administration practices and strengthened my understanding of cybersecurity fundamentals.

In the future, this project can be enhanced by adding:

- Intrusion Detection Systems (Snort/Suricata)
- Web Application Firewall (ModSecurity)
- Load Balancing
- Cloud deployment (AWS/Azure)

References

- Apache Documentation: <https://httpd.apache.org/>
- Bind9 DNS Docs: <https://bind9.readthedocs.io/>
- Postfix Docs: <http://www.postfix.org/documentation.html>
- vsftpd Docs: <https://security.appspot.com/vsftpd.html>
- Ubuntu Official Documentation
- Nmap Manual Pages
- Fail2Ban Documentation